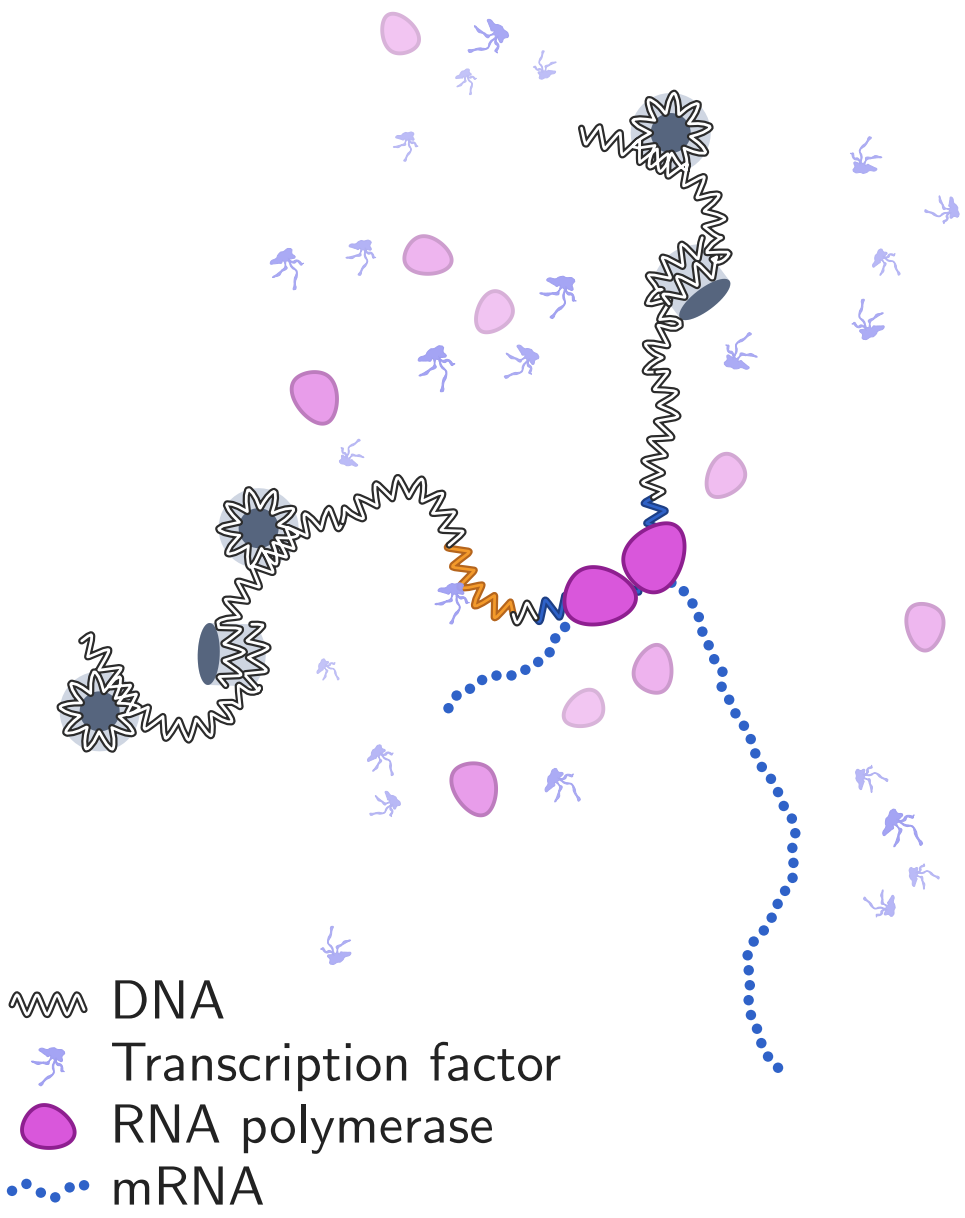
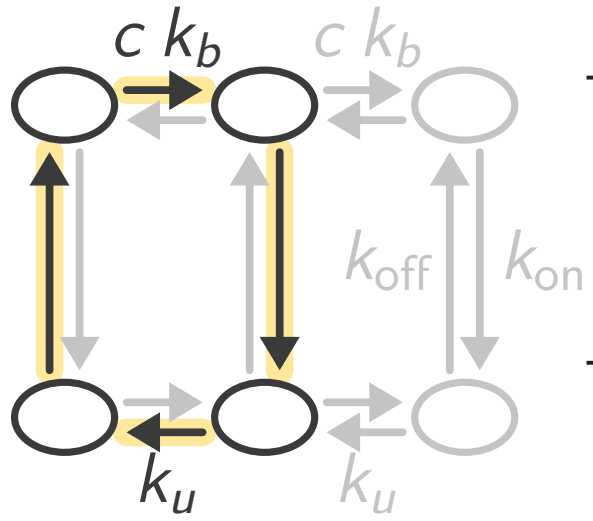
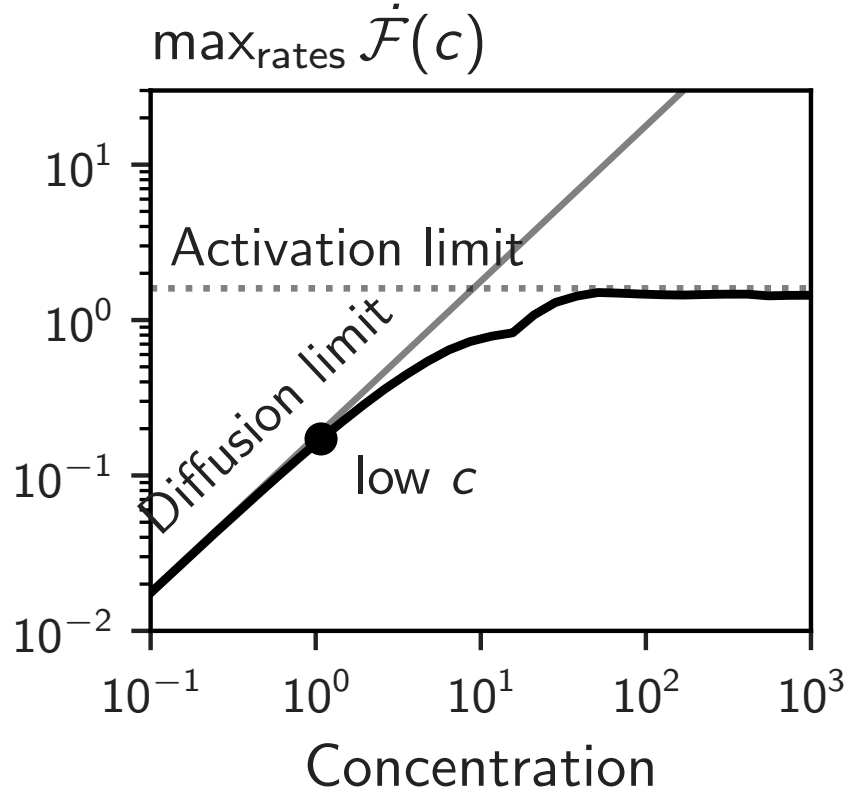

DNA
Transcription factor
RNA polymerase
mRNA


c k_b
c k_b
k_{off}
k_{on}
k_u
k_u
Transcription OFF
Transcription ON


$\max_{\text{rates}} \dot{\mathcal{F}}(c)$
Activation limit
Diffusion limit
low c
10^{1}
10^{0}
10^{-1}
10^{-2}
10^{-1}
10^{0}
10^{1}
10^{2}
10^{3}
Concentration